

How Wergonic's Numbers Work

Everything the product measures, how it decides something is dangerous, and what every number in a report means — written in plain words, with no assumed knowledge of ergonomics or statistics.

This is a reminder document. If you open a report in six months and cannot remember what “P90” or “static minutes per hour” means, section 4 and section 9 are the pages to turn to.

27 July 2026 · describes the product as it behaves today, verified against the live code

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1 The idea in one page

Sensors are strapped to a worker. They record how the body is positioned, many times a second, for the whole recording. The phone shows live feedback. Afterwards the data is uploaded, the backend works out how much time was spent in each posture, and a report gives a green, yellow or red verdict.

An angle is never dangerous by itself. It becomes dangerous when it is held for too much of the working day.

Raising your arm to 70° is fine. Raising it to 70° for three hours is not. Every risk colour in the product is really answering one question: **what share of the recording was spent in the bad position?**

That single idea explains almost everything else. The angles decide *which* bad position. The percentages decide *how bad*. The assessments (RAMP, MEC, RULA, REBA) then wrap extra judgement around it — weights being lifted, how often a motion repeats, what the worker says hurts.

The chain, start to finish

1. Sensors	Measure orientation many times a second
2. Calibration	Establishes what “neutral” means for this worker
3. Angles	Each reading becomes an angle for each body part
4. Ranges	Each angle falls into a range, e.g. 30–60°
5. Time share	How much of the recording was spent in each range
6. Colour	The time share is compared against a limit → green, yellow or red

2 What we measure

Six raw quantities. Everything in every report is derived from these.

MEASUREMENT	WHAT IT MEANS IN PLAIN WORDS	0° MEANS
Arm elevation angle	How high the upper arm is lifted	Arm hanging straight down
Arm angular velocity	How fast the arm is moving	Not moving
Trunk forward bending	Leaning forward or back from the waist	Standing upright
Trunk side bending	Leaning left or right	Upright
Head forward bending	Head tilted down or up	Head level
Heart rate	From a chest strap	—

Head side bending and wrist speed are also measured. Hand distance from the body is *calculated* rather than measured — see section 7.

How an angle is worked out

For the arm, gravity is enough. The sensor always knows which way is down. Calibration recorded which way down pointed when the arm hung by the side. During work, the sensor compares where down is now with where it was then — and the difference between those two directions *is* the elevation angle.

For the trunk and head, gravity is not enough. Here we need two separate numbers: forward bending and side bending. Gravity alone cannot tell them apart — tilting 20° forward and tilting 20° sideways shift gravity by the same amount. To know which, you need to know which way the body is facing. So for these we track full three-dimensional orientation, combining the gravity sensor with a rotation sensor.

Arms need one number, so gravity is enough. Trunk and head need two, so they need orientation.

Speed is measured in degrees per second, not metres per second

Because we measure rotation, not travel. The sensor does not know where the arm is in the room, only which way it is pointing. As a rule of thumb, **30 °/s is about the speed of raising your arm from your side to horizontal in three seconds.**

WORTH KNOWING

Arm speed is worked out two different ways in the product. The reports use the rate at which the arm rises and falls, so purely sideways movement counts as zero. The phone uses total rotation in every direction. The two will never produce the same number for the same session.

3 Calibration — why 0° means anything

A sensor on its own only knows which way gravity pulls. It has no idea where the body is. Calibration is what connects the two.

The worker stands still in a neutral pose for **six seconds**. The app records the sensor's orientation and stores it as "zero". Every angle afterwards is measured relative to that moment.

If the worker moves too much during those six seconds the calibration is rejected and they are asked to redo it. The app watches how much the readings wobble and refuses anything above a set amount.

Without calibration, every angle in the recording is meaningless.

This is the single biggest thing a user can get wrong. A sloppy calibration does not produce an error later — it produces a report full of confident, wrong numbers.

4 How we decide it's dangerous

Always two steps. This is the core of the whole product.

Step 1 — sort every moment into a range

BODY PART	RANGES
Arm elevation	up to 30° · 30–60° · above 60°
Trunk forward	Extension (past –5°) · –5 to 10° · 10–20° · 20–45° · above 45°
Head forward	Extension (past –5°) · –5 to 10° · 10–20° · 20–30° · above 30°
Side bending (trunk & head)	0–10° · 10–30° · above 30°
Arm speed	0–15 · 15–30 · above 30 °/s
Wrist speed	0–10 · 10–20 · above 20 °/s

HOW TO READ A RANGE

30° – 60° means **more than 30, up to and including 60**. The lower edge is excluded, the upper edge is included. So a reading of exactly 30.0° falls in the range below, and exactly 60.0° falls in this one. If a value sits precisely on a boundary, it goes **up** into the higher range.

DELIBERATE, NOT A MISTAKE

The range shown as “0° – 10°” for the head and trunk actually covers **–5° to 10°**. The –5° edge absorbs sensor and calibration noise, so a worker standing upright is not wrongly recorded as leaning backwards. This is intentional.

Step 2 — colour it by how much time was spent there

This table is the risk model. It lives in one file in the backend that describes itself as the single source of truth. Everything else follows from it.

RISK FACTOR	GREEN	YELLOW	RED
Arm above 60°	up to 5%	5–10%	over 10%
Arm above 30°	up to 25%	25–50%	over 50%
Trunk past 45°	up to 6%	6–12%	over 12%
Trunk past 20°	up to 12%	12–25%	over 25%
Trunk backward past –10°	up to 1%	1–6%	over 6%
Head past 30°	up to 12%	12–25%	over 25%
Head backward past –10°	up to 1%	1–6%	over 6%
Arm speed (median)	up to 15 °/s	15–30 °/s	over 30 °/s
Wrist speed (median)	up to 10 °/s	10–20 °/s	over 20 °/s

Red on “Arm above 60°” does not mean the arm went above 60°.

It means the arm was above 60° for **more than 10% of the recording**. This is the single most common misreading of our reports.

A LIMITATION TO BE AWARE OF

These are percentages of the recording, with no adjustment for how long the recording was. The published research behind these numbers gives limits **for an eight-hour workday**, and the method assumes a recording is a representative sample of the shift. If it is not — a very short recording, or one taken during an unusual task — the percentages will still be calculated but the verdict will not mean what it appears to mean.

5 Percentiles, medians and averages

Reports are full of “P50” and “P90”. They are simpler than they look.

A percentile is: sort every reading from smallest to largest, then point at a position in that line.

The sensors take roughly thirteen readings a second, so a ten-minute recording produces about eight thousand angle values. Sort all eight thousand from smallest to largest, and:

P10	One tenth of the way along. The relaxed end.
P50	The middle. Half the readings are below it. "P50" and "median" are two names for the same number.
P90	Nine tenths of the way along. One reading in ten is higher than this.
P99	Almost the very end. Near the worst moment of the recording.

Why not just use the average?

TEN READINGS OF ARM ANGLE

5, 8, 10, 12, 15, 18, 20, 25, 60, 80

Average = 25.3° — dragged upward by the two big values. Misleading: the arm was below 25° for eight of the ten readings.

Median (P50) = 16.5° — "normally the arm sits around 16° ."

P90 $\approx 60^{\circ}$ — "one tenth of the time the arm was at 60° or above."

Injury comes from sustained high exposure, and the average hides it. Two workers can have an identical average — one holding a steady 20° all day, another swinging between 0° and 40° — while their shoulders are under completely different loads. Percentiles separate them.

P50 is the posture they work in normally. P90 is the posture they reach when it gets bad.

If P50 looks fine but P90 is high, the job has bad moments inside an otherwise acceptable task. If P50 is already high, the whole task is wrong.

An important thing percentiles do not do

If the entire recording was spent in safe angles, then **even P99 will be a safe angle** — and that is the correct answer. It tells you the worker never went extreme, not even in their worst 1%.

Percentiles answer *"how high did it get?"*. The risk colours answer *"how long was it too high?"*. Two different questions, and the product uses both.

THREE WORKERS

WORKER	P50	P90	P99	TIME ABOVE 60°	VERDICT
Arm at 20° all shift	20°	22°	25°	0%	Safe
20° mostly, 15% at 80°	20°	80°	85°	15%	Dangerous
20° all shift, one 2-second reach to 100°	20°	22°	25°	0.05%	Safe

Note the third worker: **even P99 does not see the 100° reach**, because two seconds is far less than 1% of the readings. That is correct — a single brief reach is not an injury risk. Injury comes from repetition and duration.

6 What the worker sees live

During recording, each body part has a tile that changes colour. There are two feedback modes.

Instant — reacts to the current moment

BODY PART	GREY	ORANGE	RED
Arm elevation	up to 30°	30–60°	above 60°
Arm speed	0–30 °/s	30–60 °/s	above 60 °/s
Trunk / head bending	up to 20°	20–45°	above 45°
Heart rate	Always grey — never coloured during recording		

Accumulated — reacts to how the session is going so far

A second mode that builds up over the recording. For arms it turns orange once more than 25% of the time has been spent above 30° (or more than 5% above 60°), and red at more than 50% and 10%. Those are the same shapes as the risk model in section 4.

LIVE FEEDBACK AND THE REPORT DO NOT ALWAYS AGREE

The phone holds its own copies of these thresholds, and several have drifted from the ones the reports use. A worker can see green all day for a posture the report later flags. This is documented in the companion issues document — treat live feedback as guidance, and the report as the verdict.

7 The four risk assessments

These sit on top of the sensor data and answer a broader question: *how bad is this job overall?*

ASSESSMENT	AUTOMATIC OR MANUAL?	WHAT IT IS FOR
RAMP	Half and half	The most complete picture — postures, lifting, pushing, environment, and what the worker reports
MEC	Fully automatic	Repetition and holding still, measured per hour
RULA	Fully manual	Classic paper method, upper body
REBA	Fully manual	Classic paper method, whole body

RAMP

Risk Assessment and Management tool for manual handling, Proactively. Developed at KTH in Sweden. Seven sections: postures, movements and repetition, lifting, pushing and pulling, influencing factors, strenuous work, and the worker's perceived discomfort.

Postures 1 to 6 are filled in automatically from the sensors. Everything else — weights, forces, environment, what the worker says hurts — is entered by a human assessor.

How a posture question becomes a score

First we work out the share of time spent in the bad posture, then run it through a ladder:

SHARE OF TIME	SCORE
50% or more	7
37.5% or more	5
25% or more	3
12.5% or more	2
6.25% or more	1
1% or more	0.5
under 1%	0

Then the score becomes a colour: **red at 3 or above**, **yellow at 1.5 to 2**, **green below that**.

The angles RAMP uses are stricter than the general report

POSTURE	TRIGGER
1 — Head	forward past 30°, or sideways past 10°
2 — Head backwards	past -5°
3 — Back, moderate	forward 20–45°, or sideways 10–29°
4 — Back, considerable	forward past 45°, sideways past 30°, or backward past -5°
5 — Upper arm	elevation past 90°
6 — Hand distance	more than 30 cm from the body

MEC

Fully sensor-derived — no human input at all. It asks how often, and for how long, the worker sits in a demanding position. Everything is expressed *per hour*, so recordings of different lengths can be compared.

Two numbers per question:

- **Movements per hour** — how often the motion repeats
- **Static minutes per hour** — how long the position is held still

ARMS	GREEN	YELLOW	RED
Movements per hour	up to 15	15–60	over 60
Static minutes per hour	up to 4	4–8	over 8

MEC also works out **where the hand is in space** without a sensor on the hand. It treats the arm as two rigid sticks — upper arm and forearm — and uses the two measured angles plus the limb lengths to calculate how far the hand is from the body. That is how it can report “more than 30 cm” or “more than 50 cm”.

RULA and REBA

Entirely manual. No sensor input. An assessor scores each body part by eye, adds points for muscle use and load, then works through three lookup tables to reach a final score.

RULA SCORE	MEANING
1–2	Acceptable
3–4	Investigate further
5–6	Investigate and change soon
7	Investigate and change immediately

8 TNO — holding versus moving

Named after the Dutch research institute. It answers a question the other measurements do not: **was the worker holding a position, or moving through it?** Holding still and moving repeatedly cause different kinds of strain.

How it works:

1. Find the peaks and troughs in the angle signal over time
2. Group them into periods — a stretch where the angle stays within about 5°
3. A period lasting **6 seconds or more** is counted as **static** (holding); anything shorter is **dynamic** (moving)
4. Report static as a share of time, and dynamic as movements per minute — each broken down by the angle range it happened in

READ THE COMPANION ISSUES DOCUMENT BEFORE USING TNO FIGURES

A defect in the range-sorting means some angle ranges are never counted. This is the highest-priority item on the issues list and it affects every TNO figure the product has produced.

9 Reading a report, line by line

There are three different reports

REPORT	PRODUCED BY	HOW COMPLETE
Word (.docx)	The backend	The most complete. Risk verdicts, criteria text, references, appendices. This is the one to give a customer.
PDF	The web app	Similar data, different layout, seven pages
On-screen	The web app	The lightest — a working summary

They do not contain the same things. Do not expect a number in one to appear in another.

The words you will see

TERM	WHAT IT MEANS
Angle Range	A band of angles, e.g. 30–60°
Time Duration	Actual clock time spent there, as hours:minutes:seconds
Percent / Time %	Share of the recording spent there
Elevation Angle	How high the arm is lifted
Forward Bending	Leaning forward or back (trunk and head)
Side Bending	Leaning left or right. Left and right are pooled together
Extension	Leaning backwards past –5°
Velocity	Rotation speed in degrees per second
Back	Means trunk. Same body part, different word
Percentile 10 / 50 / 90 / 99	See section 5
Mean	The average
Median	The middle value — same as P50
Activity	A named task within the recording
Static minutes per hour	How long a position was held still, scaled to an hour (MEC)
Dynamic movements per hour	How often a motion repeated, scaled to an hour (MEC)

Read the duration first, then the percentage.

Every table gives you both. The colour is based on the percentage alone, so the duration is what tells you whether the percentage is meaningful.

What is in the Word report, in order

SECTION	WHAT IT HOLDS
Summary	Prose overview
High and Moderate Risk Factors	The verdict table . Only lists what came back yellow or red
Per-limb posture tables	One each for left arm, right arm, trunk, head. Angle range, duration, percent
Work Movements	Left arm, right arm and wrist speed, same three columns in %/s
Hand distance from body	Up to 30 cm, 30–50 cm, more than 50 cm, as durations
Task Percentiles	P10 / P50 / P90 / P99 broken down per activity
Session Percentiles	The same for the whole recording
Risk Assessment Criteria	The thresholds, printed out
Appendix C	Pie charts of the posture distribution
Appendix D	Methodology, definitions, references, limitations

ABOUT HAND DISTANCE

This is calculated, not measured, and it needs **both** the upper-arm and forearm sensors. If one recorded for less time than the other the estimate is incomplete — the report prints a warning with each sensor's actual duration when this happens.

The charts

CHART	ACROSS	UP	ONE POINT IS
Line chart	Time	Angle	The average angle for one whole minute
Bar chart	Angle	Share of time	How often that angle occurred
Donut chart	—	—	Share of time in each range

A TRAP IN THE LINE CHART

The line chart shows **one point per minute**, not per reading. A three-second reach overhead almost disappears into that minute's average. Use the bar chart or the percentage tables to find peaks — the line chart shows trends, not spikes.

Multi-session comparison places sessions side by side as columns. It does not merge them into one combined figure.

10 Heart rate

The heart rate block is the most jargon-heavy part of any report. All of it comes from a standard, well-established analysis library rather than our own maths.

A healthy heart does not beat like a metronome.

The tiny variations between beats show the nervous system adapting. **More variation means more recovery capacity. Less variation means more stress or fatigue.** Almost every number below is a different way of measuring that variation.

TERM	WHAT IT MEANS	DIRECTION
Mean RR	Average milliseconds between two heartbeats	Lower = working harder
Mean HR	Average beats per minute	Higher = working harder
SDNN	Overall variation between beats	Higher is better
RMSSD	Beat-to-beat variation, short term	Higher is better — recovery
NN50 / pNN50	How many consecutive beats differ by more than 50 ms	Higher = more recovery
SD1	Short-term variation	Higher is better
SD2	Long-term variation	Higher is better
SD2/SD1	Ratio of the two	Around 2–3 is typical
SampEn	How irregular and complex the rhythm is	Higher is healthier
CSI	“Fight or flight” activity	Higher = more stress
CVI	“Rest and digest” activity	Higher = better recovery

If you only remember two: RMSSD for recovery, CSI for stress.

The product also calculates **how long the worker was above a third of their heart-rate capacity** (capacity being 220 minus their age, minus their resting rate) and **how much genuine recovery time** they got — stretches of five minutes or more back near resting.

WORTH KNOWING

None of the heart-rate numbers have risk thresholds. They are shown as raw values with no colour, so a customer has no way to know whether their figure is good or bad from the report alone.

11 Where our numbers come from

The thresholds are not invented. The product cites:

SOURCE	WHAT IT GIVES US
Arvidsson et al. (2017), AMM Syd, Sweden	Arm limits and the arm velocity threshold
Hansson et al. (2016)	The eight-hour recommended limits shown as reference values
Lind et al. (2020)	Trunk limits
Rose, Lind & Forsman (2024), KTH	The RAMP II method itself
Forsman et al. (2022)	Converting IMU sensor readings into comparable posture measurements

Two deliberate deviations, both documented in the report

- 1. **Trunk backward bending.** The original RAMP method uses 0°. We use **–10°**, because of sensor sensitivity and our calibration method, to avoid overstating backward-bending risk.
- 2. **Arm velocity.** The original AMM Syd method uses 60 °/s. We use **30 °/s**, because IMU sensors estimate speed more accurately than the accelerometer-only sensors the original research used.

Both are printed in the Word report where a customer can read them. That is good practice and worth preserving.

NOT TRACEABLE

The heart-rate thresholds, the recovery percentages and the TNO cut-offs have **no published source cited anywhere in the product**. They may well be sound, but we currently cannot show where they came from if asked.

12 Things worth knowing

Short answers to questions that come up repeatedly.

QUESTION

ANSWER

Do we measure head side bending?	Yes, in the same ranges as the trunk. But it has no risk band of its own — it only feeds a RAMP score
Do we measure speed for every body part?	Arms and wrist have risk bands. Trunk and head speed is recorded and shown, but no threshold is applied to it
Is the risk model configurable per customer?	No. The posture thresholds are fixed in code. MEC alone lets the assessor set a few sensitivity parameters
Does the risk depend on the worker's age or weight?	Only for heart rate, where capacity is calculated from age. Posture thresholds are the same for everyone
Which assessments use the sensors?	MEC entirely; RAMP for postures 1 to 6 only; RULA and REBA not at all
What happens to negative arm angles?	They are removed before the distribution and risk calculations, but <i>not</i> before the percentiles
Are left and right side bending separated?	No. The size of the lean is used and the direction is discarded
Why does the phone sometimes disagree with the report?	The phone keeps its own copies of the thresholds, and several have drifted. See the companion issues document

COMPANION DOCUMENT

Calculations — Possible Issues to Confirm lists twenty places where the product may disagree with itself, each with the facts, the case for and against it being a defect, and how to settle it. This reference document describes how things behave today; that one describes what may need to change.

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Verified against backend `staging`, mobile `main` and web `dev`. Describes actual behaviour, including behaviour that may be under review. Update this document whenever a threshold changes.